

AMR GOMAA

Applied Machine Learning Researcher & PhD Candidate

✉ amr.gomaa@dfki.de
in amrgomaaelhady

📍 Saarbrücken, Germany
🔗 amrgomaaelhady

🌐 amrgomaaelhady.github.io



RESEARCH EXPERIENCE

PhD Candidate, Researcher

Deutsches Forschungszentrum für Künstliche Intelligenz, German Research Center for Artificial Intelligence (DFKI)

📅 March 2020 – March 2024 📍 Saarbrücken, Germany

- I am broadly interested in research areas related to the intersection of **machine learning** (language and vision) with **Human Computer Interaction**.
- My research topics include **multimodal gesture recognition**, **multimodal machine teaching** in Automotive and Robotics context.
- I have worked on multiple interdisciplinary industry-oriented projects with industrial partners such as **BMW** and **Carl Zeiss**.

Project Manager, TeachTAM Project

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

📅 March 2022 – March 2024 📍 Saarbrücken, Germany

- I created a project idea and established its desired goals, outcomes, milestones and acquired research funding from the German Federal Ministry of Education and Research (BMBF).
- I was responsible for leading a team of researchers in terms of coordination, organization, supervision and technical support to ensure project goals fulfillment in a timely manner.
- I coordinated and reported the results to project stakeholders (e.g. industry partners and the BMBF).

Junior Researcher

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

📅 April 2019 – March 2020 📍 Saarbrücken, Germany

- Developed and Implemented human-machine interfaces for multimodal interaction with autonomous systems (vehicle, robot) including Hardware integration
- Training end-to-end machine learning (ML) systems for such interfaces including data collections and evaluations

Student Research Assistant

HCI Group, Saarland University

📅 Oct. 2017 – April 2019 📍 Saarbrücken, Germany

- Worked on wearable devices including prototyping and novel circuit design including printed electronics, on-skin circuits and micro-controller programming
- Acknowledgements:
 - “Tacttoo: A Thin and Feel-Through Tattoo for On-Skin Tactile Output Output”, Withana et al.
 - “Tactlets: Adding Tactile Feedback to 3D Objects Using Custom Printed Controls”, Groeger et al.

EDUCATION

Ph.D. in Computer Science

Deutsches Forschungszentrum für Künstliche Intelligenz and Saarland University

📅 March. 2020 – March. 2024

- Advisor: Dr. Michael Feld
- Supervisor: Prof. Dr. Antonio Krüger

M.Sc. in Computer Science

Saarland University, Germany

📅 Oct. 2017 – August 2019

- GPA: 1.4/1.0
- Courses include: Machine Learning and Deep learning, Statistical Analysis, Human Computer Interaction, Image Processing and Computer vision, Embedded Systems and Embedded Security
- Thesis title: Investigating Person-Specific Hand Pointing and Eye Gaze Gestures for Referencing from a Moving Vehicle using Multimodal Fusion
- Advisor: Dr. Michael Feld
- Supervisor: Prof. Dr. Antonio Krüger

B.Sc. in Electrical and Computer Engineering

Ain Shams University, Egypt

📅 Aug. 2008 – Aug. 2013

- Grade: Very good with Honor (out of: pass, good, very good, excellent)
- Thesis title: Automatic Handover Optimization in 4G: LTE Self Optimizing Network (LTE-SON)

SKILLS

Machine Learning

Deep Learning

Computer Vision

Graph Neural Network

Multimodal Interaction

Python

Reinforcement Learning

PyTorch

TensorFlow

Keras

R

C

C++

Matlab

Simulink

CCNA

VBA

SELECTED PUBLICATIONS

- Amr Gomaa, et al. **Studying Person-Specific Pointing and Gaze Behavior for Multimodal Referencing of Outside Objects from a Moving Vehicle.** @ ICMI '20.
- Amr Gomaa, et al. **ML-PersRef: A Machine Learning-based Personalized Multimodal Fusion Approach for Referencing Outside Objects From a Moving Vehicle.** @ ICMI '21.
- Amr Gomaa, et al. **What's on your mind? A Mental and Perceptual Load Estimation Framework towards Adaptive In-vehicle Interaction while Driving.** @ AutoUI '22.
- Amr Gomaa, et al. **Adaptive User-Centered Multimodal Interaction towards Reliable and Trusted Automotive Interfaces.** @ ICMI '22.

INDUSTRY EXPERIENCE

Senior Radio Frequency (RF) Optimization Engineer

Vodafone Egypt

📅 May 2016 – September 2017 📍 Cairo, Egypt

- Responsible for the RF design, developments, optimization, and RF engineering services of Alexandria (the second largest city in Egypt and a fifth of Egypt's Entire Cellular Network). Also responsible for network design, development, roll out, integration and implementation of 2G/3G/4G wireless network systems.
- Worked with systems and IC designers to optimize power optimization for all RF circuitry in the cellular system. Manage multi-functional teams to optimize RF dependencies while driving the Key Radio Features.

Vodafone Discover Graduate Program

Vodafone Egypt

📅 March 2015 – April 2016 📍 Cairo, Egypt

- Discover is a global accelerated development program that enables talented fresh graduates to develop and grow within Vodafone, and ultimately become Vodafone's future leaders through; customized job rotations, challenging business exposure, continuous development, and corporate networking.
- Successfully joined a highly selective program to jump-start my career and fill a senior engineer position in a record time of only one year where I have worked with two highly experienced teams (RF Planning/Optimization & RAN Optimization).

TEACHING EXPERIENCE

Lecturer

Saarland University

📅 2019 - Ongoing 📍 Saarbrücken, Germany

Co-Lecturing several courses during my PhD Candidacy: Adaptive Human Machine Interfaces for Autonomous Systems, Hybrid Machine Learning Approaches and Applications Seminar, Automotive User Interfaces Seminar, and Machine Learning School.

Tutor

Saarland University

📅 2017 - 2019 📍 Saarbrücken, Germany

Tutored several courses during my masters studies: Human Computer Interaction and Neural Networks: Implementation and Application.

ACADEMIC SERVICES

Reviewer

📅 2020 - Ongoing

Reviewer for multiple HCI, AI and Automotive conferences such as: CHI 2023, IEEE VR 2023, NordiCHI 2022, AutomotiveUI 2021 & 2022, CHI PLAY 2021, ICMI 2021, and IEEE AIVR 2021.

Student Volunteer

📅 2019 - Ongoing

Assisted organizers in UPLINX Machine Learning Summer School 2019, CIXSchool 2022, and AutomotiveUI 2022.

SCHOLARSHIPS AND AWARDS

- Awarded the German Federal Ministry of Education and Research (**BMBF**) **research grant (100,000€)** - Software Campus Program (2021), TeachTAM Project
- Awarded the **Saarland scholarship** for international students at the University of Saarland - DAAD STIBET III scholarship grant (2019)
- Successful Selection and Participation in various Hackathons and Summer Schools such as the Sixth Summer School on Computational Interaction (**CIXSchool 2022**), the First Inria-DFKI European Summer School on AI (**IDESSAI 2021**), and **Continental's Industry 4.0 & AI Hackathon** in the Deep Learning Track (2019)
- Successful Completion of Coursera's **Google Data Analytics** Professional Certification by Google on Coursera (including 8 courses), Entire **Reinforcement Learning Specialization** (including 4 courses), Practical Reinforcement Learning course with Honors, and **Embedded Hardware and Operating Systems** course

LANGUAGES

English 

German 

REFERENCES

Prof. Dr. Antonio Krüger

@ antonio.krueger@dfki.de

✉ Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

Dr. Michael Feld

@ michael.feld@dfki.de

✉ Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)